

Course Syllabus

General Information

Faculty	Information Technology		
Department	Software Engineering		
Semester	1 st	Academic Year	2025/2026

Course Details

Course Title	Software Architecture			Course No.		0443501	
Type of Learning	☒ On- campus ☐ Blended (..... Platform) ☐ Online (..... Platform)						
Credit Hours	3	Theoretical	3	Practical	0		
Course Level (according to JNQF)			7	JNQF Hours*		12	
Pre-requisite	0442502- Software Analysis and Design.			Co-requisite		0443502 - Software Architecture lab	

Course Instructor Information

Instructor Name	Dr. Shadi Ettantawi		
Instructor E-mail	s.ettantawi@meu.edu.jo		
Course Time	8:00 – 9:00 Monday & Wednesday		
Office Hours	11:00 – 12:00 Wednesday		
Office Number	N143	Office Phone	
Lab Instructor Name –If assigned-			

Sources and References

1. Textbook

Fundamentals of Software Architecture, Mark Richards & Neal Ford, O'Reilly Media, Inc, 2020.

2. References

1. Clean Architecture A CRAFTSMAN'S GUIDE TO SOFTWARE STRUCTURE AND DESIGN, Robert C. Martin, Pearson Education, 2018.

Teaching Methods

- | | | |
|----------------|---------------------------|-------------------------------|
| 1. Lectures | 2. Case Studies | 3. Interactive demonstrations |
| 4. Assignments | 5. Discussion & Questions | |

Course Description

This course offers a comprehensive understanding of software architecture and design principles, techniques, and best practices for building robust, scalable, systems. Students will explore architectural styles while engaging in lectures, labs, and projects to develop problem solving skills and effective team work. Throughout the course, students will analyze real-world software systems, understand design decisions and trade-offs, and ultimately become equipped to tackle diverse software design challenges and create innovative and maintainable solutions.

Course Learning Outcomes (CLOs) and its correlation with Program Learning Outcomes (PLOs) and JNQF Descriptors

No.	CLOs	PLOs	JNQF Descriptors		
			Knowledge	Skill	Competency
1	Describe various software architectures concepts and design alternatives.	PLO1	K1-I		
2	Identify different design principles and implementation strategies used in software architecture and design.	PLO1	K1-I		
3	Assess modern software architecture styles with respect to real-world systems.	PLO2		S2-P	
4	Design and present a suitable software architecture for a given system within specific requirements.	PLO2		S2-D	

Course Topics

Week	Topic	Lecture Type		CLOs	Assessment	
		Lecture Type	Lecture No.		Assessment Tool	Summative/ Formative
1	1. Introduction - Defining Software Architecture - Laws of Software Architecture - Architecture Versus Design - C4 model	Lecture (1)	Lecture On- campus	1	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	1	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	1	Discussion & Questions	Formative
2	2. Modularity - Cohesion - Coupling - From Modules to Components	Lecture (1)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	2	Discussion & Questions	Formative
3	3. Architecture Characteristics 4. Architecture Styles - Unitary Architecture - Client/Server	Lecture (1)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	3	Discussion & Questions	Formative
4	- thin vs. thick clients - request-response cycle - Desktop + database server - Browser + web server - Three-tier - Monolithic Versus Distributed Architectures	Lecture (1)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	3	Discussion & Questions	Formative
5	5. Layered Architecture Style - Topology - Layers of Isolation - n-Tier & 3-Tiered systems - pros & cons - Relation to MVC	Lecture (1)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	3	Quiz	Summative
6	6. Pipeline Architecture Style - Topology - Pipes - Filters	Lecture (1)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	3	Discussion & Questions	Formative
7	7. Microkernel Architecture Style - Topology - Core System - Plug-In Components	Lecture (1)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	3	Discussion & Questions	Formative

	- Registry Contracts					
8	Mid Exam					Summative
9	8. Service-Based Architecture Style - Topology - Service Design - Database Partitioning	Lecture (1)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	3	Discussion & Questions	Formative
10	9. Event-Driven Architecture Style - Topology - Producers, consumers, and brokers.	Lecture (1)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	3	Discussion & Questions	Formative
11	10. Microservices Architecture - Topology - API Layer - Reuse - Communication	Lecture (1)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	3	Discussion & Questions	Formative
12	11. Service-Oriented Architecture (SOA) SOA vs. microservices	Lecture (1)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	3	Discussion & Questions	Formative
13	Project presentation	Lecture (1)	Lecture On- campus	3, 4	Discussion & Questions	Summative
		Lecture (2)	Lecture On- campus	3, 4	Discussion & Questions	Summative
		Lecture (3)	Lecture On- campus	3, 4	Quiz	Summative
14	12. Peer-to-Peer (P2P) architecture centralized vs. decentralized; blockchain, BitTorrent	Lecture (1)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	3	Discussion & Questions	Formative
15	13. Diagramming and Presenting Architecture - Diagramming Standards: UML, C4, and ArchiMate	Lecture (1)	Lecture On- campus	1, 4	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	1, 4	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	1, 4	Discussion & Questions	Formative
16	Final Exam	Lecture (1)	Lecture On- campus	1-4		Summative

Course Evaluation Time and Weight

Assessment Tool	Weight %	Expected Due Date
Mid Exam	25%	Week 8
Quiz	10%	Week 5, 13
Assignments	10%	Week 11
Project	15%	Week 13
Final Exam	40%	Week 16

Course Policies

Item	Policy
Quizzes	- No makeup quizzes. - A minimum of 2 quizzes are given. - Each Quiz is out of 5. - If more quizzes are given, then the lowest quiz's grade is dropped.
Exams	The format of the exams is generally (but NOT always) as follows: General Definitions, Multiple-Choice, True/False, Analyze a Problem, Essay Questions, etc.
Makeup Exams	Makeup exam should not be given unless there is a valid excuse.
Drop Date	The last day to drop the course is according to the University regulations.
Academic Honesty	- Cheating or copying on exams or quizzes is an illegal and unethical activity. - Standard MEU policy will be applied. - All graded assignments must be your own work (your own words).
Attendance	- Excellent attendance is expected. - MEU policy requires the faculty member to assign ZERO grade if a student misses 20% of the classes. - Sign-in sheets will be circulated. - If you miss class, it is your responsibility to find out about any announcements or assignments you may have missed.
E-Learning	It is your responsibility to check the course's eLearning website regularly (at least once every 2-3 days) for announcements and material.
Workload	The student should expect to spend 6 hours per week as an average workload.
Graded Exams	Instructor should return exam papers graded to students within one week after the exam date.
Participation	- Participation in and contribution to class discussions will affect your final grade positively. Raise your hand if you have any questions. - Making any kind of disruption and (side talks) in the class will affect you negatively.
Assignment/ Projects	- Assignments are as mentioned in the previous table. - Students organize themselves in teams of (2-3) students each. Each team will give at least one presentation regarding a selected case, delivering a fully working application. - More details about phases and topics of the projects will be given later on.

Others

Rubrics and Feedbacks (If Applicable)

* JNQF Hours Calculation

Activity	Duration/ week	Times/ Semester	No. of Hours
Learning Activities			
Lectures and Seminars	3	13.5	40.5
Tutorials	4	5	20
laboratory	0	0	0
Activities for Assessment and Evaluation			
First Exam/ Midterm	1	1	1
Second Exam (If applicable)			
Final Exam	2	1	2
Quizzes	1/2	2	1
Self-Learning Activities			
Assignments/ Homework	3	5	15
Case Studies	3	1	3
Presentations	3	1	3
E-learning			
Extra Readings	3	12	36
Total Hours	121.5		
JNQF Hours	Total Hours/10 = 12.15		